TRANSFORM PROPERTIES OF THE RIEMANN ZETA FUNCTION: CRITICAL-LINE PHASE COORDINATES, QUARTER-TURN GEOMETRY, EXACT LENGTH ISOMETRY, AND MONOTONE AREA ACTION

VÍCTOR CASAÑA-GINER

Abstract. For $1/2 \le \sigma \le 1$ consider the two-real-variable zeta transform

$$\mathcal{Z}(\sigma, t) = \zeta(\sigma + it).$$

The elementary Cauchy–Riemann identity $\mathcal{Z}_\sigma = -i\mathcal{Z}_t$ has a rigid geometric content. We collect and prove its exact consequences, beginning from the critical-line Hardy representation and the natural zero-to-zero phase coordinate. The two coordinate derivatives are orthogonal and isometric; the pullback metric is conformal; arbitrary oblique sections obey an exact length law; longitudinal and transverse length measures and all corresponding L^p energies coincide; the Jacobian equals the common quadratic energy density; the map is harmonic; and every t-derivative satisfies the same quarter-turn equation. For arcs whose moving endpoints remain on a fixed line through the origin, the oriented line action has derivative exactly equal to the positive Dirichlet energy, so its increment equals the oriented area swept by the transform. We also record the fixed phase-coordinate generator attached to a critical-line zero interval and its exact coefficient derivative. For the orthogonal–cubic quotient we prove nonvanishing of every projected state, projective immersion, unavoidable forcing of $\sigma = 1/2$ by the Violet node class, and global projective injectivity of the full family. No hypothesis on the location or statistics of the nontrivial zeros is used in the transform identities themselves.

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Date: Version 7 — August 2026.
ORCID: 0000-0002-8252-2608.

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1. The vertical zeta family and the critical-line trace

Definition 1.1 (Vertical zeta family)**.** *For*

$$\frac{1}{2} \le \sigma \le 1,$$

define

$$\boxed{\mathcal{Z}(\sigma, t) = \zeta(\sigma + it).} \tag{1.1}$$

The critical-line member is

$$\boxed{\mathcal{Z}_0(t) := \mathcal{Z}\left(\frac{1}{2}, t\right) = \zeta\left(\frac{1}{2} + it\right).} \tag{1.2}$$

The classical Hardy representation is

$$\boxed{\mathcal{Z}_0(t) = Z(t)e^{-i\vartheta(t)}, \qquad Z(t) \in \mathbb{R}.} \tag{1.3}$$

Thus the associated three-dimensional trace is

$$\Gamma(t) = \big(t, \Re\mathcal{Z}_0(t), \Im\mathcal{Z}_0(t)\big) = \big(t, Z(t)\cos\vartheta(t), -Z(t)\sin\vartheta(t)\big). \tag{1.4}$$

Differentiating gives

$$\boxed{\partial_t \mathcal{Z}_0(t) = e^{-i\vartheta(t)}\big(Z'(t) - i\vartheta'(t)Z(t)\big),} \tag{1.5}$$

and therefore

$$\boxed{|\partial_t \mathcal{Z}_0(t)|^2 = Z'(t)^2 + \vartheta'(t)^2 Z(t)^2.} \tag{1.6}$$

Proof. Equation (1.5) follows by the product rule, and taking the squared modulus gives (1.6) because Z, Z', ϑ' are real. □

2. Zero-to-zero intervals and the uniform phase coordinate

Let $\gamma_n < \gamma_{n+1}$ be consecutive ordinates at which

$$\mathcal{Z}_0(\gamma_n) = \mathcal{Z}_0(\gamma_{n+1}) = 0.$$

On the open interval $I_n = (\gamma_n, \gamma_{n+1})$, the real function Z has constant sign. Put

$$\varepsilon_n = \operatorname{sgn} Z(t), \qquad t \in I_n,$$

and define

$$\Omega_n = \vartheta(\gamma_{n+1}) - \vartheta(\gamma_n). \tag{2.1}$$

After the fixed rotation

$$\widehat{\mathcal{Z}}_{0,n}(t) = \varepsilon_n e^{i\vartheta(\gamma_n)}\mathcal{Z}_0(t),$$

one has

$$\widehat{\mathcal{Z}}_{0,n}(t) = |Z(t)|e^{-i[\vartheta(t)-\vartheta(\gamma_n)]}. \tag{2.2}$$

Lemma 2.1 (Global validity of the phase coordinate)**.** *For every pair $\gamma_n < \gamma_{n+1}$ of consecutive positive ordinates of zeros of the critical-line function $\mathcal{Z}_0(t)$,*

$$\boxed{\vartheta'(t) > 0 \qquad (\gamma_n \le t \le \gamma_{n+1}),} \tag{2.3}$$

and therefore

$$\boxed{\Omega_n > 0.} \tag{2.4}$$

Consequently (2.9) *below defines a unique smooth increasing diffeomorphism*

$$t_n : [0,1] \longrightarrow [\gamma_n, \gamma_{n+1}],$$

and

$$\boxed{t_n'(q) = \frac{\Omega_n}{\vartheta'(t_n(q))} > 0.} \tag{2.5}$$

Proof. From the defining formula for the Riemann–Siegel phase,

$$\vartheta(t) = \Im \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2}\log \pi,$$

one obtains

$$\vartheta'(t) = \frac{1}{2}\Re\psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2}\log \pi \tag{2.6}$$

and

$$\vartheta''(t) = -\frac{1}{4}\Im\psi'\left(\frac{1}{4} + \frac{it}{2}\right). \tag{2.7}$$

For $\Re z > 0$ the trigamma series is absolutely convergent,

$$\psi'(z) = \sum_{k=0}^{\infty} \frac{1}{(k+z)^2}.$$

Putting $z = \frac{1}{4} + \frac{it}{2}$ gives, for $t > 0$,

$$\Im \frac{1}{(k + \frac{1}{4} + \frac{it}{2})^2} = -\frac{(k+\frac{1}{4})t}{\left((k+\frac{1}{4})^2 + t^2/4\right)^2} < 0.$$

Termwise summation is legitimate by absolute convergence, hence

$$\Im\psi'\left(\frac{1}{4} + \frac{it}{2}\right) < 0, \qquad \boxed{\vartheta''(t) > 0 \quad (t > 0).}$$

Thus ϑ' is strictly increasing on $(0, \infty)$.

It remains only to locate its sign before the first positive critical-line zero. At $t = 14$, the absolutely convergent digamma representation gives

$$\psi(z) = -\gamma + \sum_{k=0}^{\infty}\left(\frac{1}{k+1} - \frac{1}{k+z}\right), \qquad \Re z > 0,$$

At $z = \frac{1}{4} + 7i$, write

$$T_k = \frac{1}{k+1} - \frac{k+\frac{1}{4}}{(k+\frac{1}{4})^2 + 49}.$$

A direct simplification gives

$$T_k = \frac{\frac{781}{16} - \frac{3k}{4}}{(k+1)((k+\frac{1}{4})^2 + 49)}.$$

For $k \geq 66$ one has $|T_k| \leq 3/(4k^2)$, and therefore, for $N \geq 66$,

$$\left|\sum_{k=N}^{\infty} T_k\right| \leq \frac{3}{4}\sum_{k=N}^{\infty}\frac{1}{k^2} \leq \frac{3}{4(N-1)}.$$

Taking $N = 200$ and using directed bounds for the finite sum, γ and $\log\pi$ yields the rigorous enclosure

$$-\gamma + \sum_{k=0}^{199} T_k > 1.9488313744452, \qquad \left|\sum_{k=200}^{\infty} T_k\right| < 0.003768844222,$$

and hence

$$\boxed{\vartheta'(14) > 0.40016 > 0.}$$

The first positive critical-line zero satisfies

$$\gamma_1 = 14.134725141734\ldots > 14.$$

Since ϑ' is strictly increasing,

$$\vartheta'(t) > \vartheta'(14) > 0 \qquad (t \geq \gamma_1).$$

Every interval $[\gamma_n, \gamma_{n+1}]$ under consideration lies in this range, proving (2.3). Hence ϑ is strictly increasing on the interval and

$$\Omega_n = \vartheta(\gamma_{n+1}) - \vartheta(\gamma_n) > 0,$$

which proves (2.4).

Therefore the restriction of ϑ to $[\gamma_n, \gamma_{n+1}]$ is one-to-one with nonvanishing derivative. Its inverse is smooth, and

$$t_n(q) = \vartheta^{-1}(\vartheta(\gamma_n) + \Omega_n q)$$

is the unique smooth increasing map onto $[\gamma_n, \gamma_{n+1}]$. Differentiating its defining identity gives

$$\vartheta'(t_n(q))t_n'(q) = \Omega_n,$$

which yields (2.5). □

Definition 2.2 (Uniform phase coordinate)**.** *Define $q \in [0,1]$ by*

$$q = \frac{\vartheta(t) - \vartheta(\gamma_n)}{\Omega_n}, \tag{2.8}$$

or equivalently define $t_n(q)$ by

$$\boxed{\vartheta(t_n(q)) = \vartheta(\gamma_n) + \Omega_n q.} \tag{2.9}$$

With

$$r_n(q) = |Z(t_n(q))|,$$

the rotated critical-line arc becomes

$$\boxed{\widehat{\mathcal{Z}}_{0,n}(q) = r_n(q)e^{-i\Omega_n q}, \qquad r_n(q) > 0 \ (0 < q < 1), \quad r_n(0) = r_n(1) = 0.} \tag{2.10}$$

Thus every zero-to-zero interval has exactly linear phase in its own natural coordinate.

For the complete vertical family, keep the same fixed map $q \mapsto t_n(q)$ and define

$$\mathcal{Z}_{\sigma,n}(q) = \mathcal{Z}(\sigma, t_n(q)). \tag{2.11}$$

No zero is moved or searched as σ varies.

3. Exact quarter-turn identity

Theorem 3.1 (Quarter-turn relation)**.** *On every open set avoiding the pole* $(\sigma, t) = (1, 0)$*,*

$$\boxed{\mathcal{Z}_\sigma = -i\mathcal{Z}_t.} \tag{3.1}$$

Consequently

$$\boxed{|\mathcal{Z}_\sigma| = |\mathcal{Z}_t|, \qquad \Re\Big(\overline{\mathcal{Z}_\sigma}\mathcal{Z}_t\Big) = 0.} \tag{3.2}$$

Proof. Since $\mathcal{Z}(\sigma, t) = \zeta(\sigma + it)$,

$$\mathcal{Z}_\sigma = \zeta'(\sigma + it), \qquad \mathcal{Z}_t = i\zeta'(\sigma + it).$$

Hence $\mathcal{Z}_\sigma = -i\mathcal{Z}_t$. Multiplication by $-i$ preserves modulus and rotates by a right angle, giving (3.2). □

Corollary 3.2 (Cauchy–Riemann form)**.** *Writing* $\mathcal{Z} = X + iY$*,*

$$X_\sigma = Y_t, \qquad Y_\sigma = -X_t. \tag{3.3}$$

4. The fixed generator in the phase coordinate

By Lemma 2.1, the phase map is globally valid on every positive zero-to-zero interval and

$$\vartheta'(t_n(q))t_n'(q) = \Omega_n, \qquad t_n'(q) = \frac{\Omega_n}{\vartheta'(t_n(q))} > 0. \tag{4.1}$$

Theorem 4.1 (Autonomous phase-coordinate evolution)**.** *For each fixed zero-to-zero interval,*

$$\boxed{\partial_\sigma \mathcal{Z}_{\sigma,n} = L_n \mathcal{Z}_{\sigma,n}, \qquad L_n = -ia_n(q)\partial_q, \qquad a_n(q) = \frac{\vartheta'(t_n(q))}{\Omega_n}.} \tag{4.2}$$

The coefficient a_n *is independent of* σ*.*

Proof. By (3.1), $\partial_\sigma \mathcal{Z} = -i\partial_t \mathcal{Z}$. The chain rule and (2.5) give

$$\partial_t = \frac{1}{t_n'(q)}\partial_q = \frac{\vartheta'(t_n(q))}{\Omega_n}\partial_q,$$

which is (4.2). □

Lemma 4.2 (Derivative of the transport coefficient)**.**

$$\boxed{a_n'(q) = \frac{\vartheta''(t_n(q))}{\vartheta'(t_n(q))}.} \tag{4.3}$$

Proof. Differentiate $a_n(q) = \vartheta'(t_n(q))/\Omega_n$ and substitute $t_n'(q) = \Omega_n/\vartheta'(t_n(q))$ from (2.5). □

5. Conformal metric and arbitrary sections

Theorem 5.1 (Exact conformal metric)**.** *For a tangent vector* $v = a\,\partial_\sigma + b\,\partial_t$*,*

$$\boxed{|d\mathcal{Z}(v)|^2 = |\mathcal{Z}_t|^2(a^2 + b^2).} \tag{5.1}$$

Equivalently,

$$\boxed{\mathcal{Z}^*(ds^2_{\mathbb{C}}) = |\mathcal{Z}_t|^2(d\sigma^2 + dt^2).} \tag{5.2}$$

Hence the transform is conformal at every point where $\mathcal{Z}_t \neq 0$*.*

Proof. Using (3.1),

$$d\mathcal{Z}(v) = a\mathcal{Z}_\sigma + b\mathcal{Z}_t = (b - ia)\mathcal{Z}_t.$$

Taking moduli proves (5.1) and (5.2). □

Corollary 5.2 (Arbitrary oblique section)**.** *For a C^1 curve $s \mapsto (\sigma(s), t(s))$,*

$$\boxed{\left|\frac{d}{ds}\mathcal{Z}(\sigma(s), t(s))\right| = |\mathcal{Z}_t|\sqrt{\sigma'(s)^2 + t'(s)^2}.} \tag{5.3}$$

Thus

$$L[\mathcal{Z} \circ \gamma] = \int |\mathcal{Z}_t(\sigma(s), t(s))|\sqrt{\sigma'(s)^2 + t'(s)^2}\, ds. \tag{5.4}$$

6. Exact longitudinal–transverse length isometry

For a measurable E in a fixed t-section define

$$\mathcal{L}_\parallel(\sigma; E) = \int_E |\mathcal{Z}_t(\sigma, t)|\, dt, \qquad \mathcal{L}_\perp(\sigma; E) = \int_E |\mathcal{Z}_\sigma(\sigma, t)|\, dt.$$

Theorem 6.1 (Equality of length measures)**.** *For every admissible σ and measurable E,*

$$\boxed{\mathcal{L}_\perp(\sigma; E) = \mathcal{L}_\parallel(\sigma; E).} \tag{6.1}$$

Equivalently, the two absolutely continuous length measures coincide:

$$\boxed{|\mathcal{Z}_\sigma(\sigma, t)|\, dt = |\mathcal{Z}_t(\sigma, t)|\, dt.} \tag{6.2}$$

Consequently, whenever the denominators are nonzero,

$$\boxed{\frac{\mathcal{L}_\perp(\sigma; E_1)}{\mathcal{L}_\perp(\sigma; E_2)} = \frac{\mathcal{L}_\parallel(\sigma; E_1)}{\mathcal{L}_\parallel(\sigma; E_2)}.} \tag{6.3}$$

Proof. Equation (3.2) gives equality of the densities pointwise. Integration proves (6.1); the ratio identity follows by division. □

Corollary 6.2 (L^p energy isometry)**.** *For every $p > 0$ for which the integrals exist,*

$$\boxed{\int_E |\mathcal{Z}_\sigma|^p\, dt = \int_E |\mathcal{Z}_t|^p\, dt.} \tag{6.4}$$

In particular the quadratic longitudinal and transverse energies agree.

Proof. Raise $|\mathcal{Z}_\sigma| = |\mathcal{Z}_t|$ to the power p and integrate. □

7. Infinitesimal distance between fibres and arbitrary directions

Theorem 7.1 (Infinitesimal inter-fibre L^1 distance)**.** *Let E be compact and assume $\mathcal{Z}$ is C^1 on a neighborhood of $\{\sigma\} \times E$. Then*

$$\boxed{\lim_{h \to 0} \frac{1}{|h|} \int_E |\mathcal{Z}(\sigma + h, t) - \mathcal{Z}(\sigma, t)|\, dt = \mathcal{L}_\parallel(\sigma; E).} \tag{7.1}$$

Proof. Uniform differentiability on the compact set gives

$$\mathcal{Z}(\sigma + h, t) - \mathcal{Z}(\sigma, t) = h\mathcal{Z}_\sigma(\sigma, t) + o(|h|)$$

uniformly in $t \in E$. Divide by $|h|$, take moduli, integrate, and use $|\mathcal{Z}_\sigma| = |\mathcal{Z}_t|$. □

Theorem 7.2 (Infinitesimal displacement in any parameter direction)**.** *For fixed real a, b,*

$$\boxed{\lim_{h\to 0} \frac{|\mathcal{Z}(\sigma + ha, t + hb) - \mathcal{Z}(\sigma, t)|}{|h|} = \sqrt{a^2 + b^2}\, |\mathcal{Z}_t(\sigma, t)|.} \tag{7.2}$$

On compact E the corresponding integrated limit is

$$\boxed{\lim_{h\to 0} \frac{1}{|h|} \int_E |\mathcal{Z}(\sigma + ha, t + hb) - \mathcal{Z}(\sigma, t)|\, dt = \sqrt{a^2 + b^2} \int_E |\mathcal{Z}_t|\, dt.} \tag{7.3}$$

Proof. Taylor expansion gives the first-order term $h(a\mathcal{Z}_\sigma + b\mathcal{Z}_t)$. Apply (5.1). Uniformity on compact sets gives the integrated statement. □

8. Jacobian and positive area density

Theorem 8.1 (Jacobian identity)**.** *Writing $\mathcal{Z} = X + iY$, the Jacobian of $(\sigma, t) \mapsto (X, Y)$ is*

$$\boxed{J_{\mathcal{Z}} = X_\sigma Y_t - X_t Y_\sigma = |\mathcal{Z}_t|^2 = |\mathcal{Z}_\sigma|^2 \geq 0.} \tag{8.1}$$

It is strictly positive exactly at regular points.

Proof. Use (3.3):

$$J_{\mathcal{Z}} = Y_t^2 + X_t^2 = |\mathcal{Z}_t|^2.$$

The second equality follows from (3.2). □

Thus the common quantity $|\mathcal{Z}_t|^2$ is simultaneously longitudinal energy density, transverse energy density, conformal scale squared, and oriented area density.

9. Strictly monotone line action

Fix a line through the origin,

$$\ell_\chi = e^{i\chi}\mathbb{R}.$$

Suppose $t_-(\sigma) < t_+(\sigma)$ are C^1 functions satisfying

$$\mathcal{Z}(\sigma, t_\pm(\sigma)) = \rho_\pm(\sigma) e^{i\chi}, \qquad \rho_\pm(\sigma) \in \mathbb{R}.$$

Define

$$\boxed{\mathscr{A}_\chi(\sigma) = \frac{1}{2} \operatorname{Im} \int_{t_-(\sigma)}^{t_+(\sigma)} \overline{\mathcal{Z}(\sigma, t)}\, \mathcal{Z}_t(\sigma, t)\, dt.} \tag{9.1}$$

Theorem 9.1 (Strict monotonicity of the line action)**.** *Under the preceding hypotheses,*

$$\boxed{\mathscr{A}'_\chi(\sigma) = \int_{t_-(\sigma)}^{t_+(\sigma)} |\mathcal{Z}_t(\sigma, t)|^2\, dt.} \tag{9.2}$$

Hence $\mathscr{A}'_\chi \geq 0$, and $\mathscr{A}'_\chi > 0$ whenever the arc is nonconstant.

Proof. Write $z = \mathcal{Z}$. Differentiation with moving endpoints gives

$$\mathscr{A}'_\chi = \frac{1}{2} \operatorname{Im} \int_{t_-}^{t_+} (\overline{z_\sigma} z_t + \overline{z} z_{t\sigma})\, dt + \frac{1}{2} \operatorname{Im} \left[\overline{z} z_t t'_\pm\right]_{t_-}^{t_+}.$$

Integrating the $z_{t\sigma}$ term by parts gives

$$\mathscr{A}'_\chi = \int_{t_-}^{t_+} \operatorname{Im}(\overline{z_\sigma} z_t)\, dt + \frac{1}{2} \operatorname{Im} \left[\overline{z}\, (z_\sigma + z_t t'_\pm)\right]_{t_-}^{t_+}.$$

At each endpoint, $z = \rho e^{i\chi}$ and the total derivative is $\rho' e^{i\chi}$, so $\overline{z}\, dz/d\sigma = \rho\rho' \in \mathbb{R}$ and the boundary contribution vanishes. Finally $z_\sigma = -iz_t$, hence

$$\operatorname{Im}(\overline{z_\sigma} z_t) = \operatorname{Im}(i|z_t|^2) = |z_t|^2.$$

This proves (9.2). □

Corollary 9.2 (Local continuation of transverse line cuts)**.** *If at σ_0 the transform has intersections with ℓ_χ at $t_-^0 < t_+^0$ and*

$$\partial_t \operatorname{Im}(e^{-i\chi} \mathcal{Z}(\sigma_0, t_\pm^0)) \neq 0,$$

then the implicit function theorem gives local C^1 continuations $t_\pm(\sigma)$ and (9.2) holds on that neighborhood.

Proof. Apply the implicit function theorem to

$$F(\sigma, t) = \operatorname{Im}(e^{-i\chi} \mathcal{Z}(\sigma, t)).$$

The displayed derivative is exactly $F_t(\sigma_0, t_\pm^0)$. □

10. Action, energy, and swept area

Theorem 10.1 (Exact action–energy–area identity)**.** *Let $\sigma_1 < \sigma_2$ lie on a regular branch of the moving cuts. Then*

$$\boxed{\mathscr{A}_\chi(\sigma_2) - \mathscr{A}_\chi(\sigma_1) = \int_{\sigma_1}^{\sigma_2} \int_{t_-(\sigma)}^{t_+(\sigma)} |\mathcal{Z}_t|^2 \, dt \, d\sigma = \iint_D J_{\mathcal{Z}} \, d\sigma \, dt,} \tag{10.1}$$

where D is the corresponding parameter strip between the two moving cuts.

Proof. Integrate (9.2) in σ and substitute the Jacobian identity (8.1). □

By the planar area formula, the last integral in (10.1) is the Euclidean area of the image counted with multiplicity:

$$\boxed{\iint_D J_{\mathcal{Z}}(\sigma, t) \, d\sigma \, dt = \int_{\mathbb{C}} N(\mathcal{Z}, D; w) \, dA(w),} \tag{10.2}$$

where $N(\mathcal{Z}, D; w)$ denotes the number of preimages of w in D, counted almost everywhere. Since $J_{\mathcal{Z}} = |\mathcal{Z}_t|^2 \geq 0$, there is no cancellation of orientation on the regular set. In particular, if $\mathcal{Z}$ is injective on D, then $N(\mathcal{Z}, D; w) = 1$ almost everywhere on $\mathcal{Z}(D)$ and 0 outside it, and therefore

$$\boxed{\mathscr{A}_\chi(\sigma_2) - \mathscr{A}_\chi(\sigma_1) = \operatorname{Area}(\mathcal{Z}(D)).} \tag{10.3}$$

Thus, without injectivity the action increment is area counted with multiplicity, while under injectivity it is the ordinary Euclidean area of the image.

Corollary 10.2 (No regular action cycle)**.** *If the arc is nonconstant on a set of positive parameter area, then it cannot return at $\sigma_2 > \sigma_1$ to a configuration with the same oriented line action while remaining on the same regular cut branch.*

Proof. The right-hand side of (10.1) is strictly positive. □

11. Critical-line value of the action

On a zero-to-zero interval in the uniform phase coordinate, (2.10) gives

$$\overline{\widehat{\mathcal{Z}}}_{0,n}\,\partial_q\widehat{\mathcal{Z}}_{0,n} = r_n r_n' - i\Omega_n r_n^2.$$

Hence the oriented action of the critical-line arc closed through the origin is

$$\boxed{\mathscr{A}_{0,n} = -\frac{\Omega_n}{2}\int_0^1 r_n(q)^2\,dq.} \tag{11.1}$$

This is an exact amplitude–phase formula attached to the zero interval.

12. Harmonicity and the differential hierarchy

Theorem 12.1 (Harmonicity)**.** *The transform satisfies*

$$\boxed{\mathcal{Z}_{\sigma\sigma} + \mathcal{Z}_{tt} = 0.} \tag{12.1}$$

Thus both real coordinates X and Y are harmonic in (σ, t).

Proof. Differentiate (3.1) with respect to σ:

$$\mathcal{Z}_{\sigma\sigma} = -i\mathcal{Z}_{t\sigma} = -i(-i\mathcal{Z}_{tt}) = -\mathcal{Z}_{tt}.$$

□

Theorem 12.2 (Differential hierarchy)**.** *For every integer $m \geq 0$,*

$$\boxed{\partial_\sigma(\partial_t^m\mathcal{Z}) = -i\partial_t(\partial_t^m\mathcal{Z}).} \tag{12.2}$$

Consequently every t-derivative inherits the same local conformal, length, Jacobian, and harmonic identities on its regular set.

Proof. Differentiate (3.1) m times with respect to t and commute derivatives. □

13. Flatness of the induced metric

On a simply connected regular region write

$$\lambda(\sigma, t) = \log|\mathcal{Z}_t(\sigma, t)|.$$

Theorem 13.1 (Harmonic conformal factor)**.** *Where $\mathcal{Z}_t \neq 0$,*

$$\boxed{\lambda_{\sigma\sigma} + \lambda_{tt} = 0.} \tag{13.1}$$

Consequently the conformal metric

$$ds^2 = e^{2\lambda}(d\sigma^2 + dt^2)$$

has Gaussian curvature zero.

Proof. The function $\mathcal{Z}_t$ is holomorphic as a function of $s = \sigma + it$ on the regular region, so $\log|\mathcal{Z}_t|$ is harmonic. For a conformal metric $e^{2\lambda}(dx^2 + dy^2)$, the Gaussian curvature is $-e^{-2\lambda}\Delta\lambda$, which vanishes by (13.1). □

14. Status of the orthogonal–cubic projective extension

For completeness, we record separately the present logical status of the orthogonal–cubic projective construction associated with a fixed zero-to-zero phase cell. Let

$$\mathcal{H} = L^2([0,1], dq; \mathbb{C}), \qquad \mathcal{P}_3 = \operatorname{span}\{1, q, q^2, q^3\}, \qquad Q_\perp = I - \Pi_3,$$

where Π_3 is the orthogonal projector onto $\mathcal{P}_3$, and set

$$u_{\sigma,n} = Q_\perp W_{\sigma,n}, \qquad M_n(\sigma) = 1 - \frac{|\langle u_{1/2,n}, u_{\sigma,n}\rangle|^2}{\|u_{1/2,n}\|^2 \|u_{\sigma,n}\|^2}$$

whenever the projected states are nonzero. The exact Gram-kernel identity is

$$M_n'(\sigma) = 2(1 - M_n(\sigma)) \int_{1/2}^{\sigma} \mathcal{C}_n(\alpha, \sigma)\, d\alpha,$$

with

$$\mathcal{C}_n(\sigma,\sigma) = P_n(\sigma) = \frac{\|u_{\sigma,n} \wedge u_{\sigma,n}'\|^2}{\|u_{\sigma,n}\|^4} \geq 0$$

on the regular projective set.

14.1. **Pole separation, critical forcing, and global projective rigidity.** Define the holomorphic phase extension

$$\Theta(z) = \frac{1}{2i}\left[\log\Gamma\left(\frac{1}{4} + \frac{iz}{2}\right) - \log\Gamma\left(\frac{1}{4} - \frac{iz}{2}\right)\right] - \frac{z}{2}\log\pi, \tag{14.1}$$

on the strip

$$S = \{z \in \mathbb{C} : |\Im z| < 1/2\},$$

and put

$$q_n(z) = \frac{\Theta(z) - \Theta(\gamma_n)}{\Omega_n}. \tag{14.2}$$

The two Gamma arguments in (14.1) have positive real part on S, so compatible holomorphic logarithms exist there. At the lower boundary point $z_0 = -i/2$, one has at most logarithmic growth,

$$q_n(z) = O(1 + |\log(z - z_0)|) \qquad (z \longrightarrow z_0,\ z \in S). \tag{14.3}$$

Lemma 14.1 (Pole-separation lemma)**.** *Let $1/2 \leq \alpha < \beta \leq 1$, let $\lambda \in \mathbb{C}^\times$, and let $p_3 \in \mathcal{P}_3$. Then the identity*

$$\zeta(\alpha + it_n(q)) - \lambda\zeta(\beta + it_n(q)) = p_3(q) \tag{14.4}$$

cannot hold on a nonempty real q-interval.

Proof. If (14.4) held almost everywhere, continuity would make it hold everywhere on the interval. Replacing q by the real phase coordinate and applying the identity theorem gives the meromorphic identity

$$\zeta(\alpha + iz) - \lambda\zeta(\beta + iz) = p_3(q_n(z))$$

on S. Set

$$z_\beta = -i(1 - \beta).$$

Because $\beta > \alpha \geq 1/2$, one has $\beta > 1/2$, hence $z_\beta \in S$. Moreover,

$$\beta + iz_\beta = 1, \qquad \alpha + iz_\beta = 1 - (\beta - \alpha) \neq 1.$$

Thus the second zeta term has a simple pole at z_β, whereas the first zeta term and $p_3(q_n(z))$ are holomorphic there. Since $\lambda \neq 0$, the pole cannot cancel, a contradiction. □

Proposition 14.2 (Nonvanishing of every orthogonal–cubic state)**.** *For every fixed positive phase cell and every* $1/2 \le \sigma \le 1$*,*

$$\boxed{u_{\sigma,n} = Q_\perp W_{\sigma,n} \neq 0.} \tag{14.5}$$

Proof. Suppose first that $\sigma > 1/2$ and $Q_\perp W_{\sigma,n} = 0$. Then $W_{\sigma,n} = p_3$ in L^2 for some $p_3 \in \mathcal{P}_3$; continuity upgrades this to pointwise equality. Analytic continuation gives

$$\zeta(\sigma + iz) = p_3(q_n(z))$$

on S. At $z_\sigma = -i(1-\sigma) \in S$, the left side has a simple pole and the right side is holomorphic, a contradiction.

For $\sigma = 1/2$, approach the boundary point $z_0 = -i/2$ from within S. Writing $z = z_0 + w$, one has

$$\zeta\left(\frac{1}{2} + iz\right) = \zeta(1 + iw) = \frac{1}{iw} + O(1),$$

whereas (14.3) gives

$$p_3(q_n(z)) = O\Big(1 + |\log w|^3\Big).$$

A simple pole cannot equal a cubic polynomial in a logarithm. This again gives a contradiction. □

Proposition 14.3 (Global projective immersion)**.** *For every fixed positive phase cell and every* $1/2 \le \sigma \le 1$*,*

$$\boxed{u_{\sigma,n} \wedge \partial_\sigma u_{\sigma,n} \neq 0.} \tag{14.6}$$

Consequently

$$\boxed{P_n(\sigma) = \frac{\|u_{\sigma,n} \wedge \partial_\sigma u_{\sigma,n}\|^2}{\|u_{\sigma,n}\|^4} > 0.} \tag{14.7}$$

Proof. If the wedge vanished, Proposition 14.2 would give a constant $c \in \mathbb{C}$ such that

$$Q_\perp(\partial_\sigma W_{\sigma,n} - cW_{\sigma,n}) = 0.$$

Hence, for some $p_3 \in \mathcal{P}_3$,

$$\zeta'(\sigma + it_n(q)) - c\zeta(\sigma + it_n(q)) = p_3(q). \tag{14.8}$$

For $\sigma > 1/2$, continue to $z_\sigma = -i(1-\sigma) \in S$. The ζ' term has a double pole, the zeta term has at most a simple pole, and the right side is holomorphic. The double pole cannot cancel.

For $\sigma = 1/2$, let $z = z_0 + w$ as above. Then

$$\zeta'(1 + iw) = \frac{1}{w^2} + O(1), \qquad \zeta(1 + iw) = \frac{1}{iw} + O(1),$$

up to the immaterial nonzero constant in the double-pole term, while $p_3(q_n(z)) = O(1 + |\log w|^3)$. Again the double pole is impossible. Thus the wedge never vanishes, and (14.7) follows. □

Theorem 14.4 (Unavoidable critical-line forcing by the Violet node class)**.** *For every fixed Violet zero-to-zero phase cell and every* $1/2 \le \sigma \le 1$*,*

$$\boxed{[Q_\perp W_{\sigma,n}] = [Q_\perp W_{1/2,n}] \quad \Longleftrightarrow \quad \sigma = \frac{1}{2}.} \tag{14.9}$$

Equivalently,

$$\boxed{M_n(\sigma) = 0 \quad \Longleftrightarrow \quad \sigma = \frac{1}{2}.} \tag{14.10}$$

Thus the value $1/2$ *is forced by the orthogonal–cubic Violet node class; it is not merely a freely chosen reference label.*

Proof. The reverse implication is immediate. If $\sigma > 1/2$ and the classes were equal, Proposition 14.2 would give $\lambda \in \mathbb{C}^\times$ and $p_3 \in \mathcal{P}_3$ such that

$$\zeta\left(\frac{1}{2} + it_n(q)\right) - \lambda\zeta(\sigma + it_n(q)) = p_3(q).$$

This contradicts Lemma 14.1 with $\alpha = 1/2$ and $\beta = \sigma$. Hence $\sigma = 1/2$ is unavoidable. □

Corollary 14.5 (Global projective injectivity of Rainbow)**.** *For every fixed positive phase cell and all $\alpha, \beta \in [1/2, 1]$,*

$$\boxed{[Q_\perp W_{\alpha,n}] = [Q_\perp W_{\beta,n}] \quad \Longleftrightarrow \quad \alpha = \beta.} \tag{14.11}$$

Therefore every Rainbow parameter determines a distinct orthogonal–cubic projective class.

Proof. If $\alpha \neq \beta$, interchange them if necessary and assume $\alpha < \beta$. Equality of the projective classes gives $\lambda \in \mathbb{C}^\times$ and $p_3 \in \mathcal{P}_3$ satisfying

$$\zeta(\alpha + it_n(q)) - \lambda\zeta(\beta + it_n(q)) = p_3(q),$$

contradicting Lemma 14.1. □

Theorem 14.6 (Strict initial orthogonal–cubic separation)**.** *For every fixed positive phase cell, at $\sigma_0 = 1/2$ one has*

$$\boxed{M_n(\sigma_0) = 0, \qquad M_n'(\sigma_0) = 0, \qquad M_n''(\sigma_0) = 2P_n(\sigma_0) > 0.} \tag{14.12}$$

Consequently there exists $\varepsilon_n > 0$ such that

$$\boxed{M_n'(\sigma) > 0 \qquad (1/2 < \sigma < 1/2 + \varepsilon_n).} \tag{14.13}$$

Proof. The definition gives $M_n(\sigma_0) = 0$. In the exact identity

$$M_n'(\sigma) = 2(1 - M_n(\sigma)) \int_{\sigma_0}^{\sigma} \mathcal{C}_n(\alpha, \sigma)\, d\alpha,$$

the integral vanishes at $\sigma = \sigma_0$, so $M_n'(\sigma_0) = 0$. Differentiating once more at σ_0 gives

$$M_n''(\sigma_0) = 2\mathcal{C}_n(\sigma_0, \sigma_0) = 2P_n(\sigma_0).$$

Proposition 14.3 gives $P_n(\sigma_0) > 0$. Continuity of M_n'' then implies $M_n'(\sigma) > 0$ on a sufficiently small right neighborhood of σ_0. □

14.2. **Conditional global monotonicity criterion.** The modal argument proves a complete criterion for global orthogonal–cubic monotonicity. We record it explicitly, separating the proved implication from the two estimates required to apply it unconditionally to the zeta phase-cell family. Let the exact reflection decomposition be

$$u_\sigma = \mathsf{X}_\sigma + \mathsf{Y}_\sigma,$$

where X is the symmetric core and Y is the antisymmetric remainder. Put

$$K_+(\alpha, \sigma) = \langle \mathsf{X}_\alpha, \mathsf{X}_\sigma \rangle, \qquad K(\alpha, \sigma) = \langle u_\alpha, u_\sigma \rangle,$$

$$\Delta_\sigma^{(+)} = \mathsf{X}_\sigma \wedge \mathsf{X}_\sigma', \qquad \Delta_\sigma = u_\sigma \wedge u_\sigma', \qquad E_\sigma = \Delta_\sigma - \Delta_\sigma^{(+)},$$

and

$$A_+(\alpha, \sigma) = \langle \Delta_\alpha^{(+)}, \Delta_\sigma^{(+)} \rangle.$$

Write the symmetric-core Plücker coordinates as

$$D_{jk}(\sigma) = \widetilde{b}_j(\sigma)\widetilde{b}_k'(\sigma) - \widetilde{b}_j'(\sigma)\widetilde{b}_k(\sigma),$$

where fixed modal signs have been chosen so that $\widetilde{b}_m(\sigma) > 0$ on $[1/2, 1]$. Suppose

$$D_{m,m+1}(\sigma) > 0 \qquad \left(\frac{1}{2} \le \sigma \le 1\right). \tag{14.14}$$

Then, for $j < k$,

$$\frac{D_{jk}}{\widetilde{b}_j \widetilde{b}_k} = \sum_{\ell=j}^{k-1} \left(\frac{\widetilde{b}'_{\ell+1}}{\widetilde{b}_{\ell+1}} - \frac{\widetilde{b}'_{\ell}}{\widetilde{b}_{\ell}} \right) > 0,$$

so every D_{jk} is positive. Consequently

$$A_+(\alpha, \sigma) = \sum_{j<k} D_{jk}(\alpha) D_{jk}(\sigma) > 0.$$

For a fixed adjacent pair $m_0, m_0 + 1$, put

$$d_{*,n} = \min_{\sigma \in [1/2,1]} D_{m_0,m_0+1}(\sigma) > 0, \qquad B_n = \max_{\sigma \in [1/2,1]} \|\mathsf{X}_\sigma\|.$$

Then

$$\boxed{\Re \frac{A_+(\alpha,\sigma)}{K_+(\alpha,\sigma)^2} \ge \frac{d_{*,n}^2}{B_n^4} =: \mu_n > 0.} \tag{14.15}$$

Define

$$U_0 = \sup \|\mathsf{X}_\sigma\|, \quad U_1 = \sup \|\mathsf{X}'_\sigma\|, \quad y_0 = \sup \|\mathsf{Y}_\sigma\|, \quad y_1 = \sup \|\mathsf{Y}'_\sigma\|,$$

$$k_* = \inf_{\alpha,\sigma} K_+(\alpha,\sigma) > 0, \quad \eta_* = \frac{2U_0 y_0 + y_0^2}{k_*}, \quad e_* = U_0 y_1 + y_0 U_1 + y_0 y_1,$$

$$\varepsilon_* = \frac{2 e_* U_0 U_1 + e_*^2}{k_*^2}, \qquad R_* = \frac{U_0^2 U_1^2}{k_*^2}.$$

Lemma 14.7 (Exact reflection-perturbation estimate)**.** *If $\eta_* < 1$, then $K(\alpha,\sigma) \neq 0$ and*

$$\Re \frac{\langle \Delta_\alpha, \Delta_\sigma \rangle}{K(\alpha,\sigma)^2} \ge \Re \frac{A_+(\alpha,\sigma)}{K_+(\alpha,\sigma)^2} - \frac{\varepsilon_* + R_* \eta_* (2 + \eta_*)}{(1 - \eta_*)^2}. \tag{14.16}$$

Proof. Write $\delta K = K - K_+$. Cauchy–Schwarz gives

$$|\delta K| \le 2U_0 y_0 + y_0^2 = \eta_* k_* \le \eta_* |K_+|.$$

Hence

$$|K| \ge (1 - \eta_*)|K_+| \ge (1 - \eta_*) k_* > 0.$$

Moreover,

$$E_\sigma = \mathsf{X}_\sigma \wedge \mathsf{Y}'_\sigma + \mathsf{Y}_\sigma \wedge \mathsf{X}'_\sigma + \mathsf{Y}_\sigma \wedge \mathsf{Y}'_\sigma,$$

so $\|E_\sigma\| \le e_*$. Therefore

$$|\langle \Delta_\alpha, \Delta_\sigma \rangle - A_+(\alpha,\sigma)| \le 2 e_* U_0 U_1 + e_*^2.$$

After division by $|K|^2$, this contribution is at most $\varepsilon_*/(1 - \eta_*)^2$.

Finally set $\delta = \delta K / K_+$. Since $|\delta| \le \eta_*$,

$$\left| \frac{K_+^2}{K^2} - 1 \right| = \left| \frac{1}{(1+\delta)^2} - 1 \right| \le \frac{\eta_* (2 + \eta_*)}{(1 - \eta_*)^2}.$$

Using $|A_+| \le U_0^2 U_1^2$ and $|K_+| \ge k_*$ gives the remaining error term in (14.16). Taking real parts completes the proof. □

Theorem 14.8 (Global orthogonal–cubic monotonicity criterion)**.** *Assume the fixed modal signs and* (14.14)*. If*

$$\boxed{\eta_* < 1, \qquad \varepsilon_* + R_*\eta_*(2+\eta_*) \le \mu_n(1-\eta_*)^2,} \tag{14.17}$$

then

$$\boxed{\mathcal{C}_n(\alpha,\sigma) \ge 0 \qquad \left(\frac{1}{2} \le \alpha \le \sigma \le 1\right),} \tag{14.18}$$

and consequently

$$\boxed{M_n'(\sigma) \ge 0 \qquad \left(\frac{1}{2} \le \sigma \le 1\right).} \tag{14.19}$$

Hence the orthogonal–cubic Fubini–Study separation from the critical-line fibre never decreases.

Proof. The symmetric-core estimate (14.15), together with Lemma 14.7, gives

$$\mathcal{C}_n(\alpha,\sigma) \ge \mu_n - \frac{\varepsilon_* + R_*\eta_*(2+\eta_*)}{(1-\eta_*)^2}.$$

Condition (14.17) makes the right-hand side nonnegative, proving (14.18). The exact projective derivative identity then yields

$$M_n'(\sigma) = 2(1 - M_n(\sigma)) \int_{1/2}^{\sigma} \mathcal{C}_n(\alpha,\sigma)\, d\alpha \ge 0.$$

Finally, with $M_n = \sin^2 \Theta_n$,

$$M_n' = \sin(2\Theta_n)\Theta_n',$$

so $\Theta_n' \ge 0$ wherever $0 < \Theta_n < \pi/2$, and the nondecreasing conclusion extends through the endpoints by continuity. □

Remark 14.9. *The theorem is a proved implication. Its unconditional application to the actual zeta phase-cell family requires proving* (14.14) *and* (14.17) *for that family. The earlier derivation itself identified exactly these two estimates as the remaining tasks; they are not silently assumed here.*

The critical-line forcing theorem and global projective injectivity are unconditional and exclude every repeated orthogonal–cubic projective class. Global monotonicity is stronger. The conditional argument above proves it from the explicit Chebyshev and perturbative inequalities (14.14)–(14.17); what remains is their unconditional verification for the actual zeta phase-cell family.

15. General holomorphic-transform principle

The preceding transform identities are not peculiar to the zeta function.

Theorem 15.1 (Holomorphic quarter-turn principle)**.** *Let f be holomorphic on a domain in $\mathbb{C}$ and put*

$$F(x,y) = f(x+iy).$$

Then

$$F_x = -iF_y,$$

and every theorem above that uses only this identity, differentiability, and the stated endpoint conditions holds verbatim for F.

Proof. The proof is identical: $F_x = f'(x+iy)$ and $F_y = if'(x+iy)$. □

TABLE 1. Logical status of the principal orthogonal–cubic statements.

Statement	Status
$u_{\sigma,n} \neq 0$ on the full parameter interval	proved in Proposition 14.2
$u_{\sigma,n} \wedge u'_{\sigma,n} \neq 0$ on the full parameter interval	proved in Proposition 14.3
$P_n(\sigma) > 0$ on the full parameter interval	proved in (14.7)
Strict positivity of $\mathcal{C}_n(\alpha, \sigma)$ in a neighborhood of the diagonal	proved locally by continuity from $P_n(\sigma) > 0$
$M'_n(\sigma) > 0$ immediately to the right of $\sigma = 1/2$	proved in Theorem 14.6
$M_n(\sigma) = 0 \iff \sigma = 1/2$	proved unconditionally by Theorem 14.4
$[Q_\perp W_{\alpha,n}] = [Q_\perp W_{\beta,n}] \iff \alpha = \beta$	proved unconditionally by Corollary 14.5
No return to the initial orthogonal–cubic projective class	proved, equivalent to the preceding uniqueness statement
No projective stationary point	proved from $u_{\sigma,n} \wedge u'_{\sigma,n} \neq 0$
$M'_n(\sigma) \geq 0$ for every $1/2 \leq \sigma \leq 1$	proved under Theorem 14.8; two zeta-specific estimates remain to be verified unconditionally
Global TP2: $\mathcal{C}_n(\alpha, \sigma) \geq 0$ on the full parameter triangle	proved under the same explicit criterion; unconditional verification for the zeta family remains open

16. Logical separation of universal and zeta-specific structure

The quarter-turn geometry, conformal metric, length isometries, Jacobian identity, harmonicity, differential hierarchy, and line-action law are universal consequences of holomorphy. The critical-line representation (1.3), the zero-to-zero intervals, the phase coordinate (2.9), and the explicit amplitude–phase action (11.1) are the additional zeta-specific structures used here. None of the transform theorems requires the Riemann hypothesis, pair correlation, a zero-density hypothesis, or a statistical model for the zeros.

The equality of longitudinal and transverse lengths is a statement at the same value of σ. It does not assert that longitudinal lengths at two distinct vertical fibres are scalar multiples of one another.

17. Information, zeros, and the exact scope of monotonicity

In the half-plane $\Re s > 1$ one has

$$\zeta(s) = \sum_{m \geq 1} m^{-s},$$

whose Dirichlet coefficients are all equal to 1 and therefore positive. On $1/2 \leq \Re s \leq 1$ the ordinary Dirichlet series does not converge; the transform there is defined by analytic continuation, not by a new convergent positive-coefficient Dirichlet series.

The coordinate map $s = \sigma + it$ and the phase reparametrization $t = t_n(q)$ are invertible and lose no information. Hence zeros are preserved exactly before projection:

$$W_{\sigma,n}(q) = 0 \quad \Longleftrightarrow \quad \zeta(\sigma + it_n(q)) = 0.$$

The quotient $Q_\perp$, however, removes the cubic component, so zeros are not invariants of an arbitrary class in $\mathcal{H}/\mathcal{P}_3$. Projectivization additionally removes a nonzero complex scale. What is proved here is the stronger restricted statement that these losses do not identify two members of the Rainbow family:

$$\sigma \longmapsto [Q_\perp W_{\sigma,n}]$$

is injective. Thus σ and consequently the original fibre and its zeros are recoverable on the Rainbow image, although an arbitrary L^2 function and its discarded cubic component are not recoverable from the quotient class.

Finally, three different sign statements must be kept separate. The Jacobian identity

$$J_{\mathcal{Z}} = |\mathcal{Z}_t|^2 \geq 0$$

is unconditional on the full regular transform domain. The line action is strictly increasing on every persistent regular branch of fixed-line cuts. The global orthogonal–cubic inequality $M_n'(\sigma) \geq 0$ is proved by Theorem 14.8 under the explicit Chebyshev and perturbative estimates; their verification for the actual zeta phase-cell family remains the stated global sign problem.

References

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[2] L. V. Ahlfors, *Complex Analysis*, 3rd ed., McGraw–Hill, 1979.

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